

Probability

GCSE Maths · Practice Questions

20 questions · 62 marks total

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Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1 [1 mark]

A fair six-sided dice is rolled once.
Write down the probability of rolling a 3.

.....

2 [2 marks]

A bag contains 5 red counters, 3 blue counters and 2 green counters.
One counter is taken at random.
(a) Write down the probability that it is blue.
(b) Write down the probability that it is not red.

.....

3 [2 marks]

The probability that it rains tomorrow is 0.35
Work out the probability that it does not rain tomorrow.

.....

4 [2 marks]

A spinner has four colours. The table shows some of the probabilities.

Colour	Red	Blue	Green	Yellow
Probability	0.2	0.3	0.15	x

Work out the value of x .

.....

5

[3 marks]

- (a) A fair six-sided dice is rolled 300 times. Estimate how many times it will land on 4.
- (b) A biased coin has $P(\text{head}) = 0.4$. It is tossed 250 times. Estimate the number of heads.
-

6

[3 marks]

60 students travel to school by bus or on foot. The two-way table is incomplete.

	Walk	Bus	Total
Boys	15		35
Girls			
Total	24		60

A student is chosen at random. Work out the probability that the student is a girl who travels by bus.

7

[3 marks]

Two fair six-sided dice are rolled and their scores are added together.

- (a) Work out the probability that the total is 7.
- (b) Work out the probability that the total is greater than 9.
-

8

[3 marks]

A drawing pin is dropped 200 times. It lands point up 130 times.

- (a) Estimate the probability that the pin lands point up.
- (b) The pin is going to be dropped 500 times. Estimate how many times it will land point up.
-

9

[4 marks]

In a group of 40 students, 22 study French, 18 study German and 9 study both languages.

- (a) How many students study neither language?
- (b) One student is chosen at random. Work out the probability that they study French but not German.
- (c) Work out the probability that they study at least one of the two languages.

.....

10

[4 marks]

A bag contains 4 red counters and 6 blue counters.

A counter is taken at random, its colour recorded, and it is **put back**. A second counter is then taken.

- (a) Work out the probability that both counters are red.
- (b) Work out the probability that one counter is red and the other is blue.

.....

Section B · Higher

Higher-tier style. Again, questions 1–3 are the gentlest and question 10 is the hardest.

1

[2 marks]

A and B are independent events. $P(A) = 0.3$ and $P(B) = 0.5$

Work out $P(A \text{ and } B)$.

.....

2

[2 marks]

A bag contains 5 red and 7 blue counters. One counter is taken and **not** replaced. A second counter is then taken.

The first counter was red. Write down the probability that the second counter is also red.

.....

3

[3 marks]

A bag contains 3 green and 5 yellow sweets.

Two sweets are taken at random without replacement.

Work out the probability that both are green.

.....

4

[3 marks]

The probability that a machine breaks down on any given day is 0.05, independently of other days.
Work out the probability that the machine breaks down on at least one of three days.

5

[4 marks]

60% of the students at a school travel in by bus.
Of those who travel by bus, 15% are late. Of those who do not travel by bus, 5% are late.
(a) Work out the probability that a student chosen at random is late.
(b) A student is late. Work out the probability that this student came by bus.

6

[4 marks]

A bag contains 4 white counters and 6 black counters.
Two counters are taken at random without replacement.
Work out the probability that the two counters are the same colour.

7

[4 marks]

100 people were asked whether they own a car (set A) and whether they own a bicycle (set B).
45 own a car, 38 own a bicycle and 20 own both.
One person is chosen at random. Work out
(a) $P(A \cup B)$
(b) $P(A' \cap B')$
(c) $P(A | B)$

8

[4 marks]

A bag contains n red counters and 5 blue counters.
Two counters are taken at random without replacement.
The probability that both counters are red is $\frac{7}{22}$

- (a) Show that $3n^2 - 17n - 28 = 0$
(b) Work out the number of red counters in the bag.
-

9

[4 marks]

In a group of 30 students, 18 like maths, 14 like science and 6 like neither subject.
A student who likes maths is chosen at random.
Work out the probability that this student also likes science.

10

[5 marks]

A box contains 3 red sweets and 7 green sweets.
Three sweets are taken at random without replacement.
Work out the probability that at least one of them is red.
